

Quasielastic lepton-nucleus scattering with *ab initio* spectral functions from nuclear matter

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Alma Cavallin, 17/09-2026

[arXiv:2608.23057](https://arxiv.org/abs/2608.23057)

Modeling quasielastic lepton-nucleus interactions with *ab initio* spectral functions from infinite nuclear matter

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(Dated: 24th August 2026)

We present a study of quasielastic lepton-nucleus scattering within the local density approximation, using *ab initio* spectral functions from infinite nuclear matter derived with self-consistent Green's functions theory and interpolated by neural networks. We include final-state interactions in terms of the particle spectral function and investigate the importance and range of validity of this treatment for varying momentum transfer. The performance of the model is tested for inclusive electron scattering on different isospin-symmetric target nuclei, and we present results for ^{12}C , ^{16}O , and ^{40}Ca . Theoretical uncertainties originating from the neural network interpolation and the Hamiltonian dependence are assessed. We also present a calculation of charged-current responses of ^{16}O and the flux-averaged total ν_μ - ^{12}C cross section. Our model shows good agreement with experimental data within the range of validity of our approximations. The framework can be readily extended to include additional dynamical mechanisms, such as pion production, as well as other nuclear Hamiltonians.

Work in collaboration with:

Joanna E. Sobczyk (Chalmers)

Francesco Marino (U. Mainz)

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[arXiv:2601.03763](https://arxiv.org/abs/2601.03763)

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Gorkov algebraic diagrammatic construction for infinite nuclear matter

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We propose a novel many-body truncation for Gorkov self-consistent Green's function (SCGF) theory where pairing correlations are handled at first order, while dynamical correlations are described using the particle-number-conserving Dyson-SCGF scheme up to third order in the algebraic diagrammatic construction. The new method is enabled by the introduction of a scheme that allows us to approximate the Gorkov propagator in terms of a particle-number-conserving optimized reference state. The approach provides state-of-the-art predictions of the equation of state and spectral properties of infinite nuclear matter at zero temperature and in the presence of pairing. We find satisfactory results using modern saturating Hamiltonians at next-to-next-to-leading order in chiral effective field theory.

DOI: [10.1103/r4j1-ggmr](https://doi.org/10.1103/r4j1-ggmr)

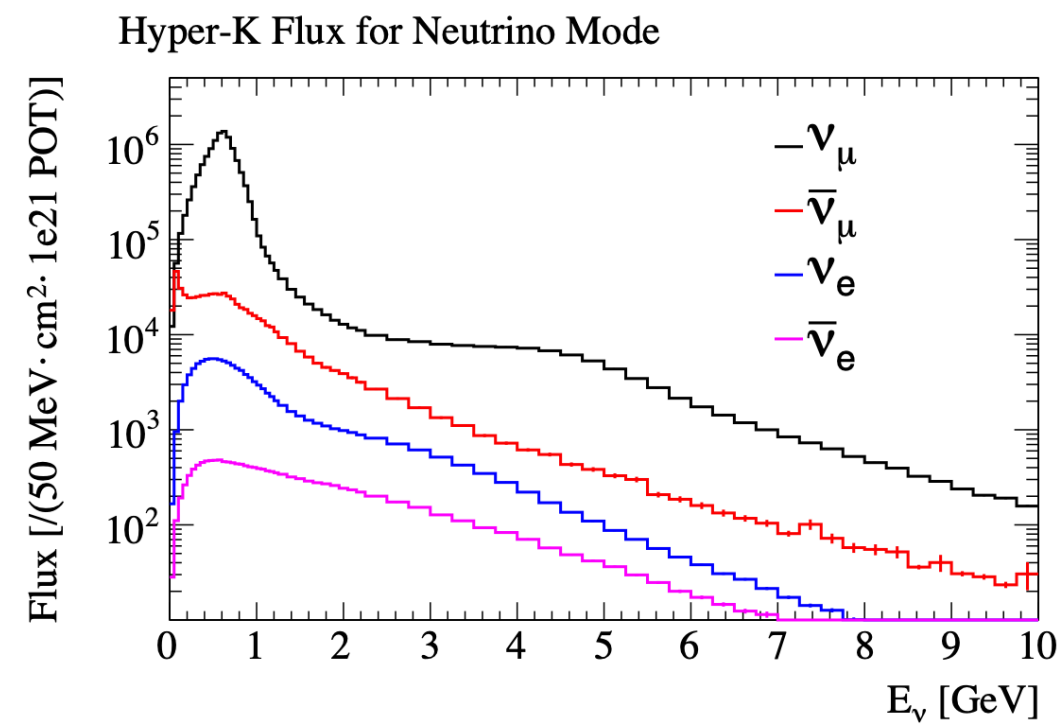
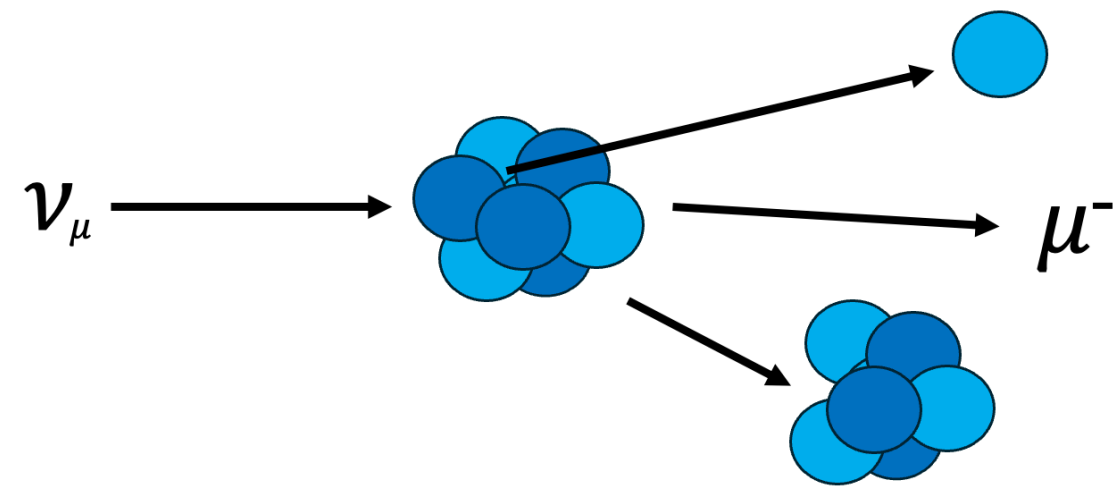
Work in collaboration with:

Joanna E. Sobczyk (Chalmers)

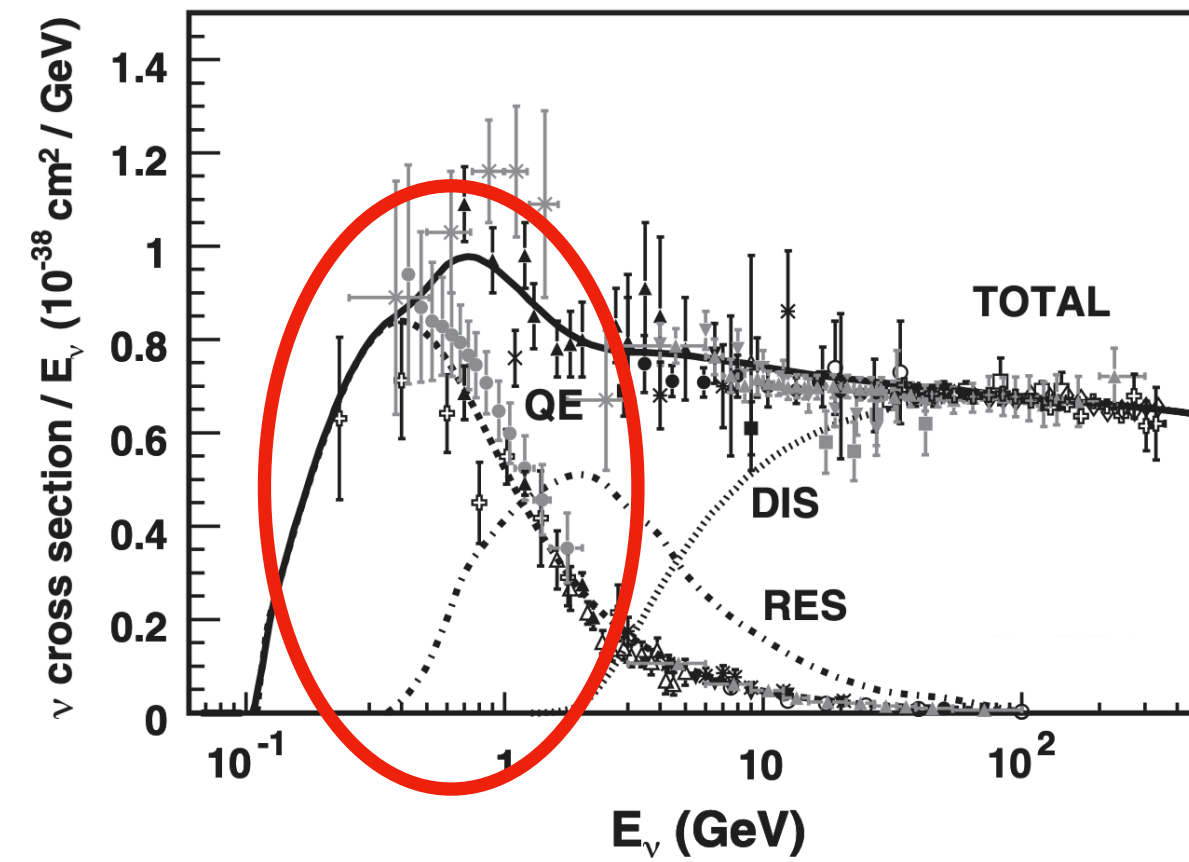
Francesco Marino (U. Mainz)

Quasielastic lepton-nucleus scattering with *ab initio* spectral functions from nuclear matter

Dominant process in T2HK and DUNE



K. Abe *et al.* (2018)

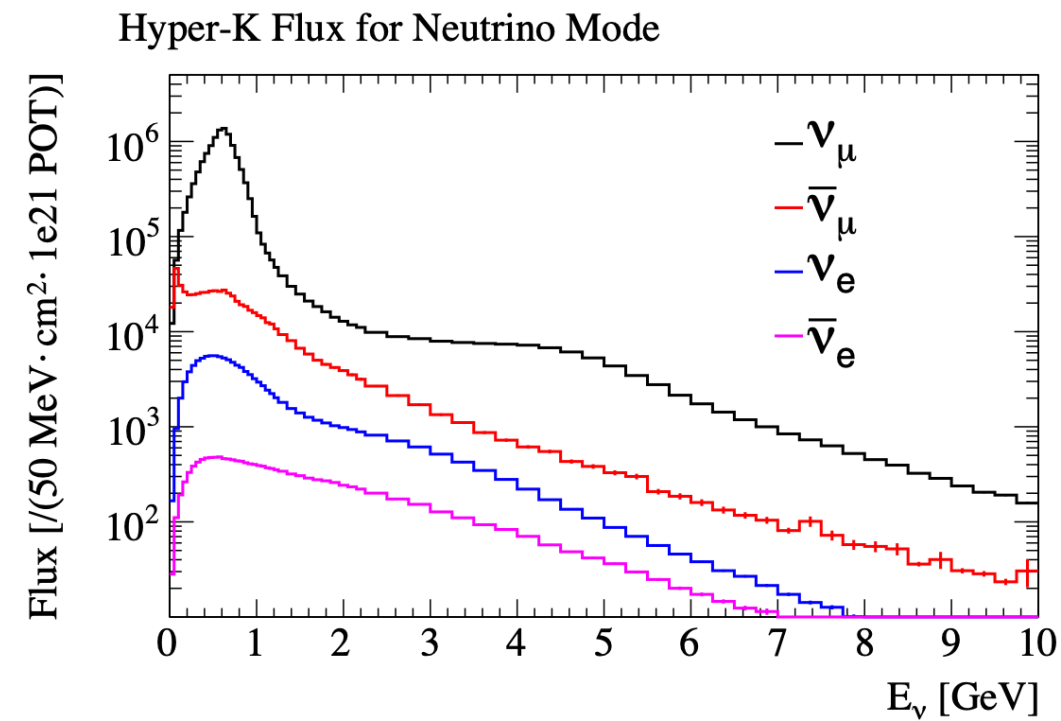
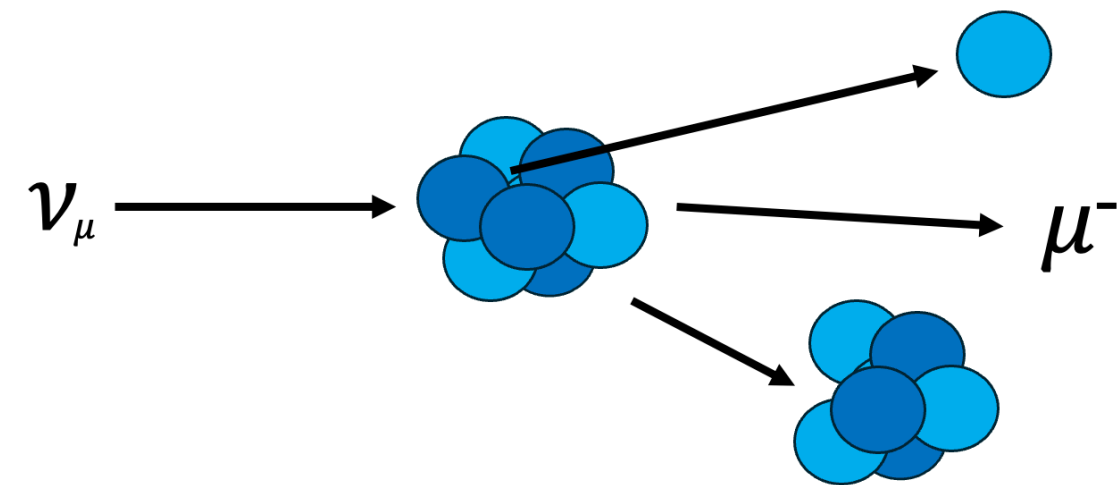


J. A. Formaggio *et al.* (2012)

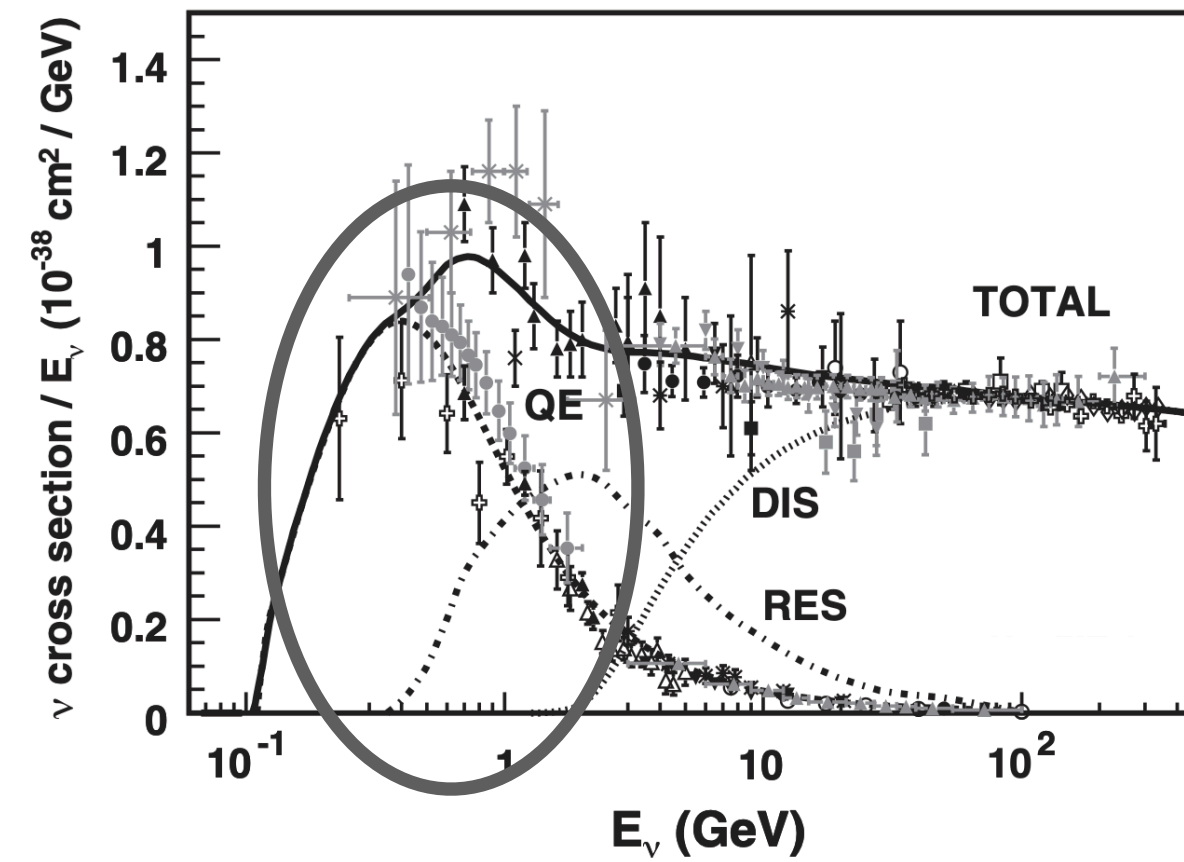
Quasielastic lepton-nucleus scattering with *ab initio* spectral functions from nuclear matter

Dominant process in T2HK and DUNE

Quantify and reduce theoretical uncertainties



K. Abe *et al.* (2018)

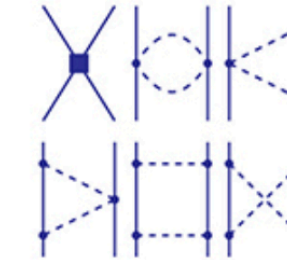


J. A. Formaggio *et al.* (2012)

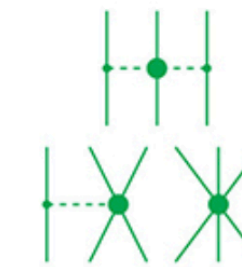
LO
(Q/Λ_χ)⁰



NLO
(Q/Λ_χ)²



NNLO
(Q/Λ_χ)³

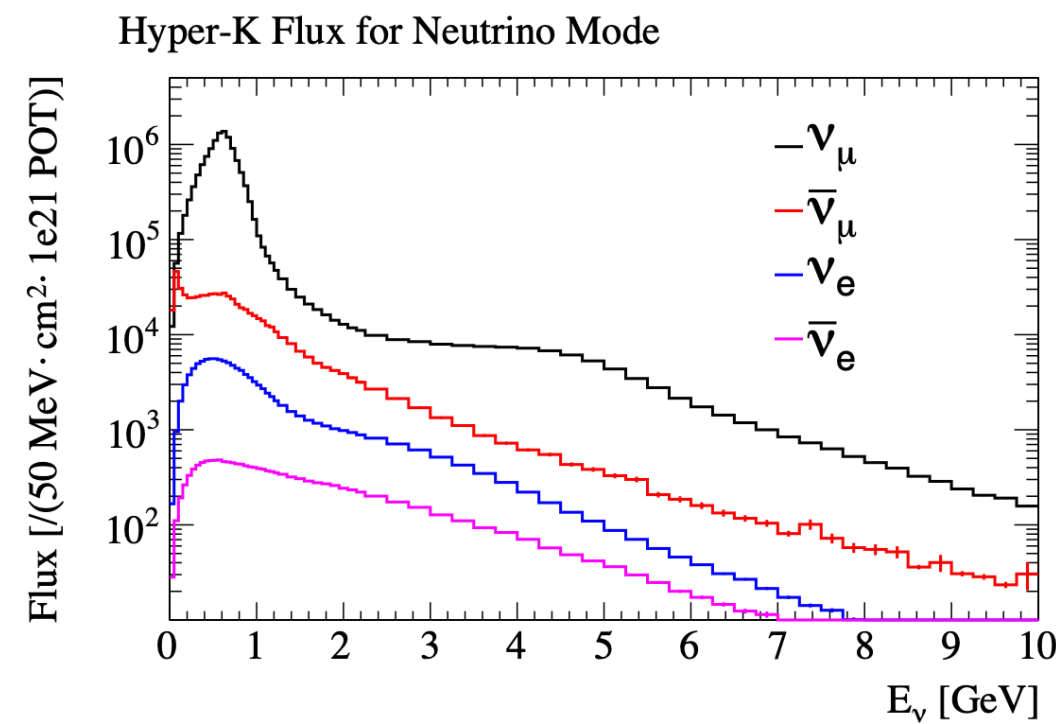
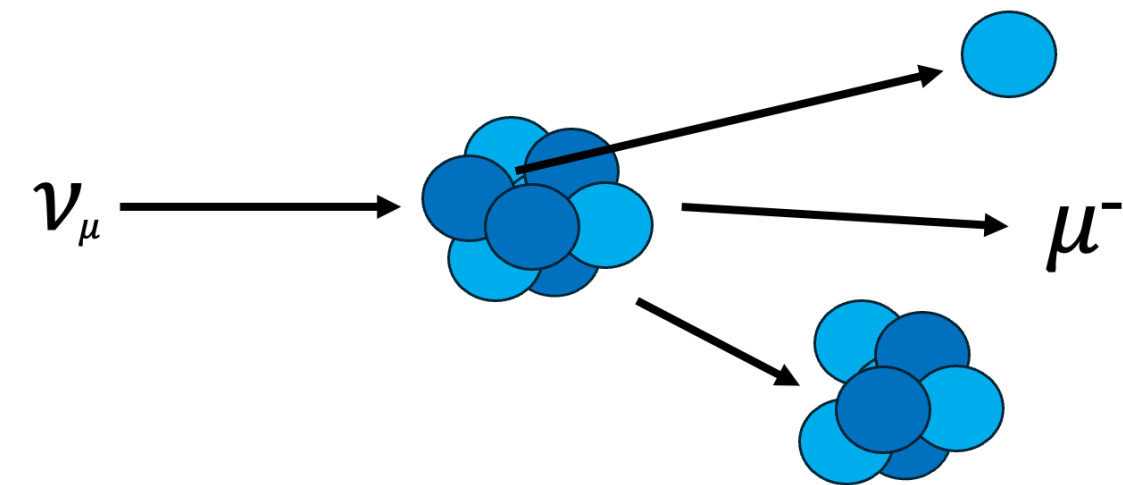


Machleidt *et al.* (2011)

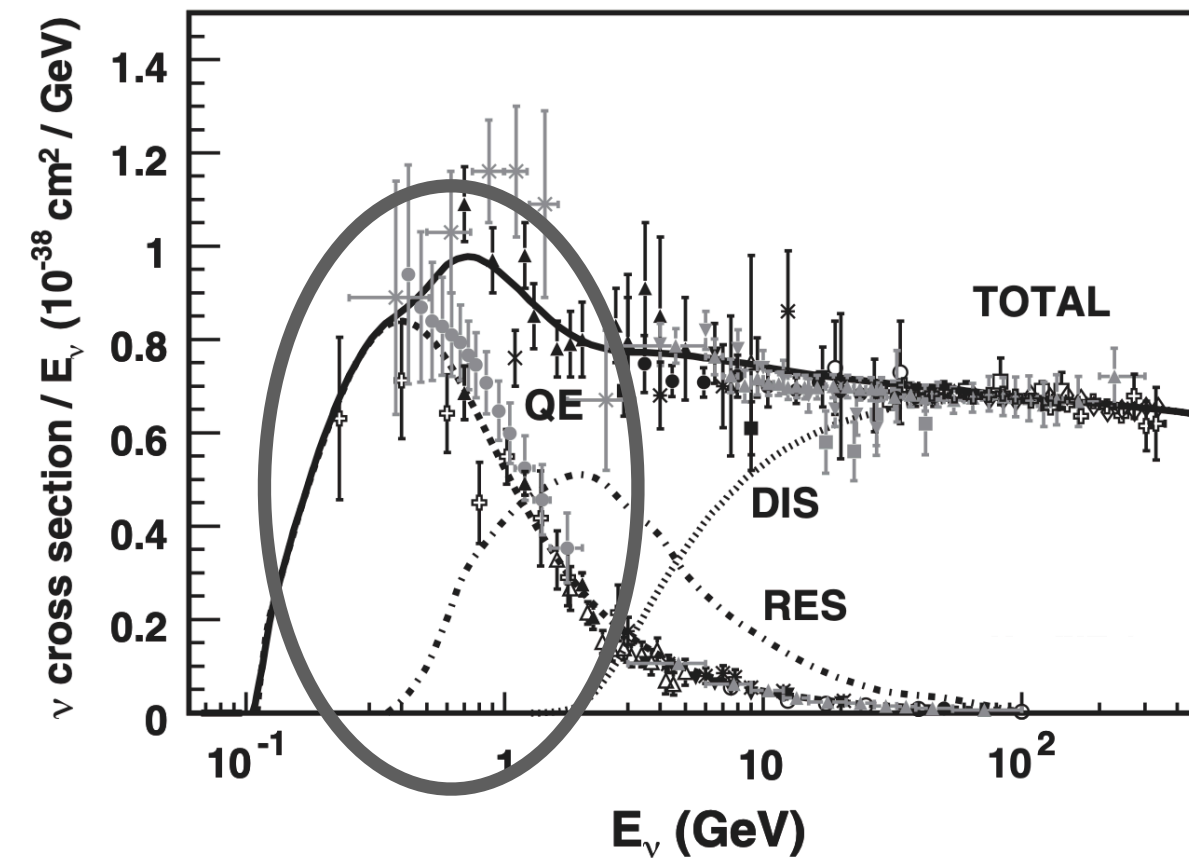
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Dominant process in T2HK and DUNE

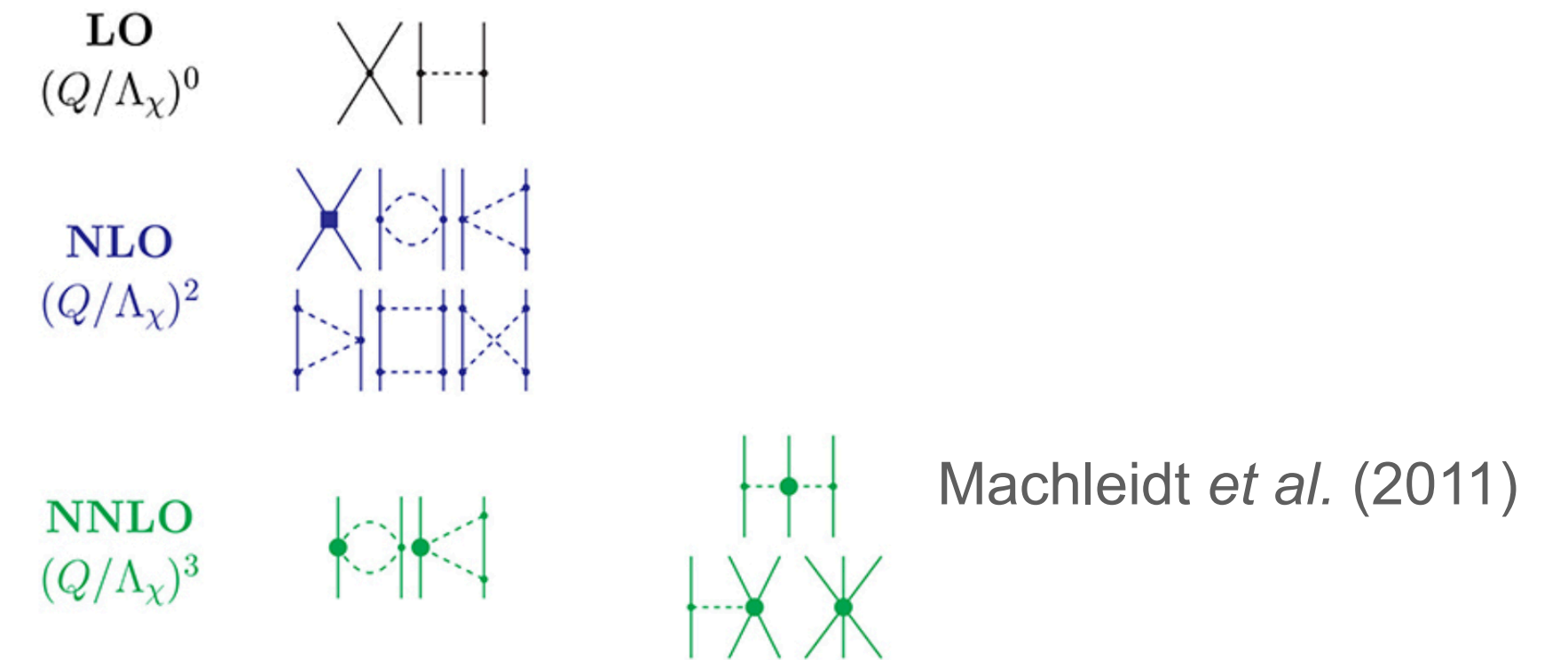
Quantify and reduce theoretical uncertainties



K. Abe *et al.* (2018)



J. A. Formaggio *et al.* (2012)



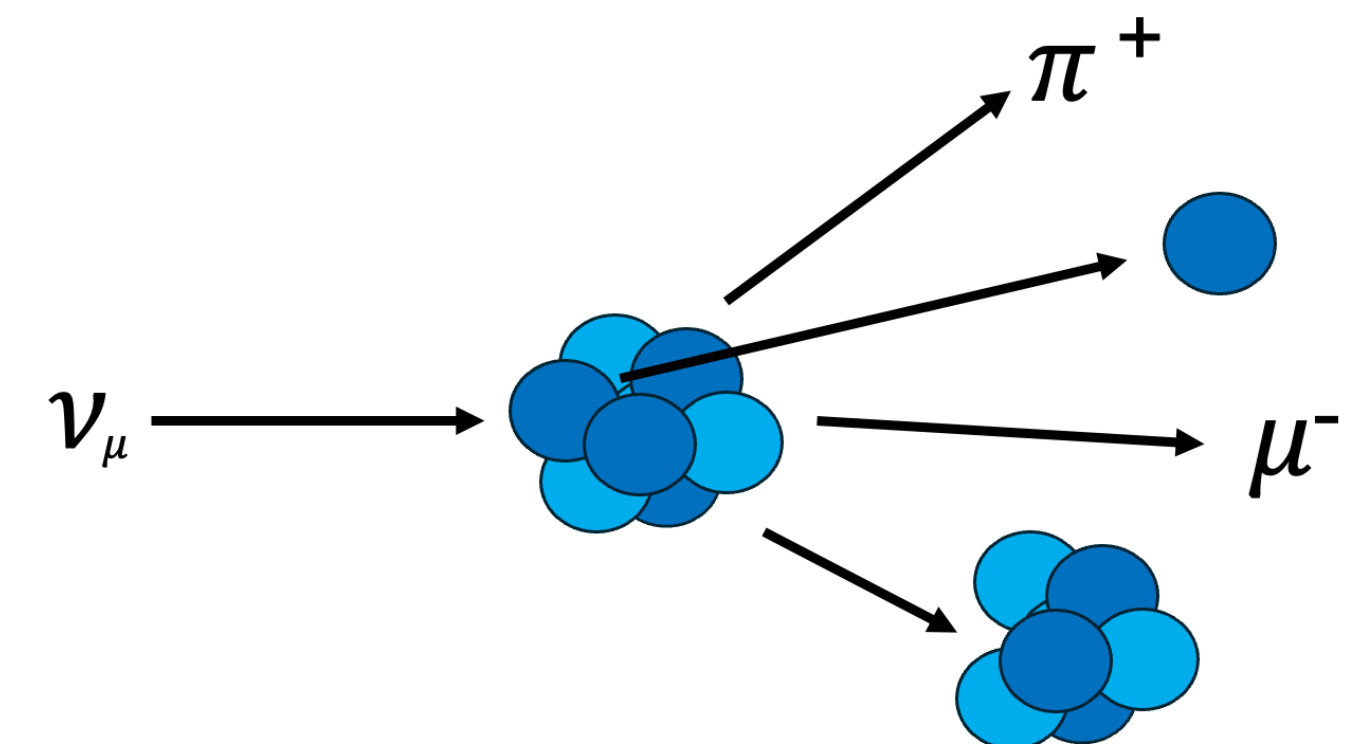
Extended to additional mechanisms

LDA suitable for MC event generators

Different target nuclei within the same framework

Relativistic currents

$$j^\mu = F_1 \gamma^\mu + i \frac{F_2}{2m_N} \sigma^{\mu\nu} q_\nu - F_A \gamma^\mu \gamma^5 - \frac{F_P}{m_N} \gamma^5 q^\mu$$



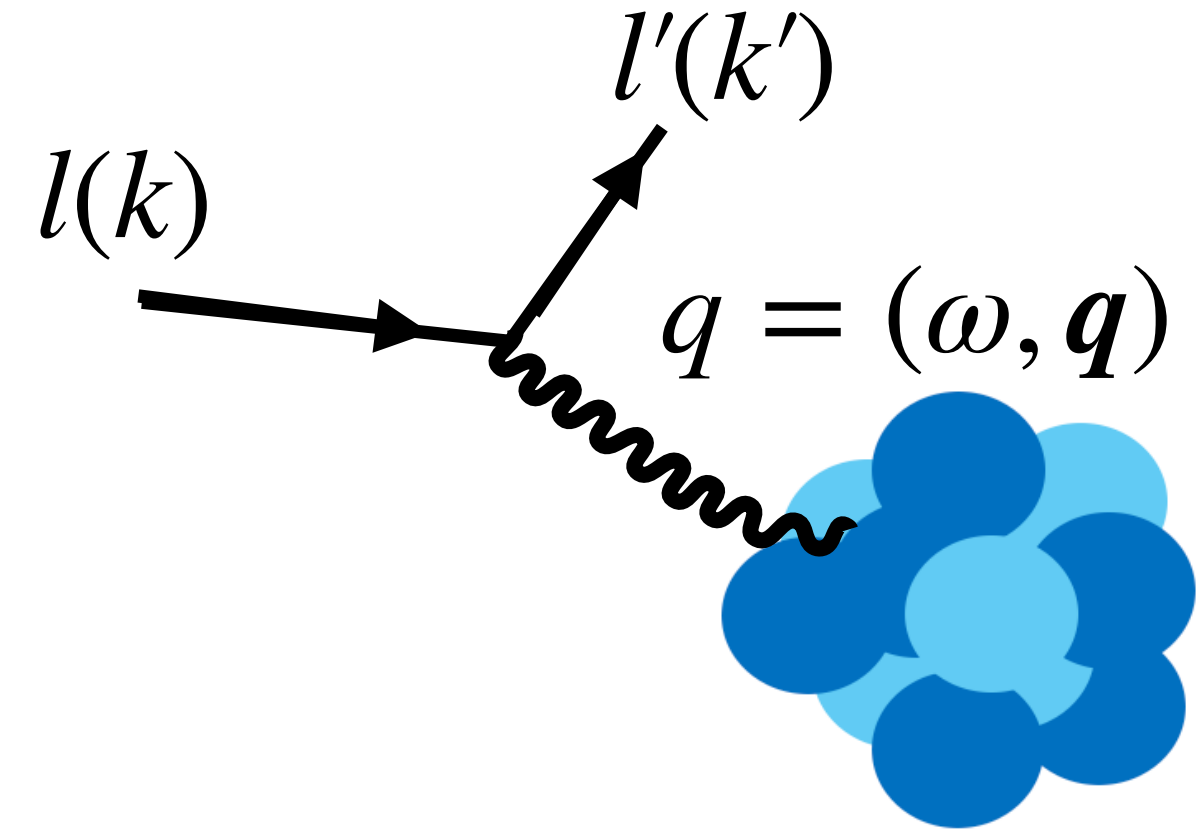
Outline

- **Theoretical formalism**
Ab initio final-state interactions
- **Neural-network interpolation of *ab initio* spectral functions**
- **Results**
Electroweak response functions, inclusive electron and neutrino cross sections
- **Summary and next steps**

Theoretical formalism

Inclusive lepton-nucleus cross section

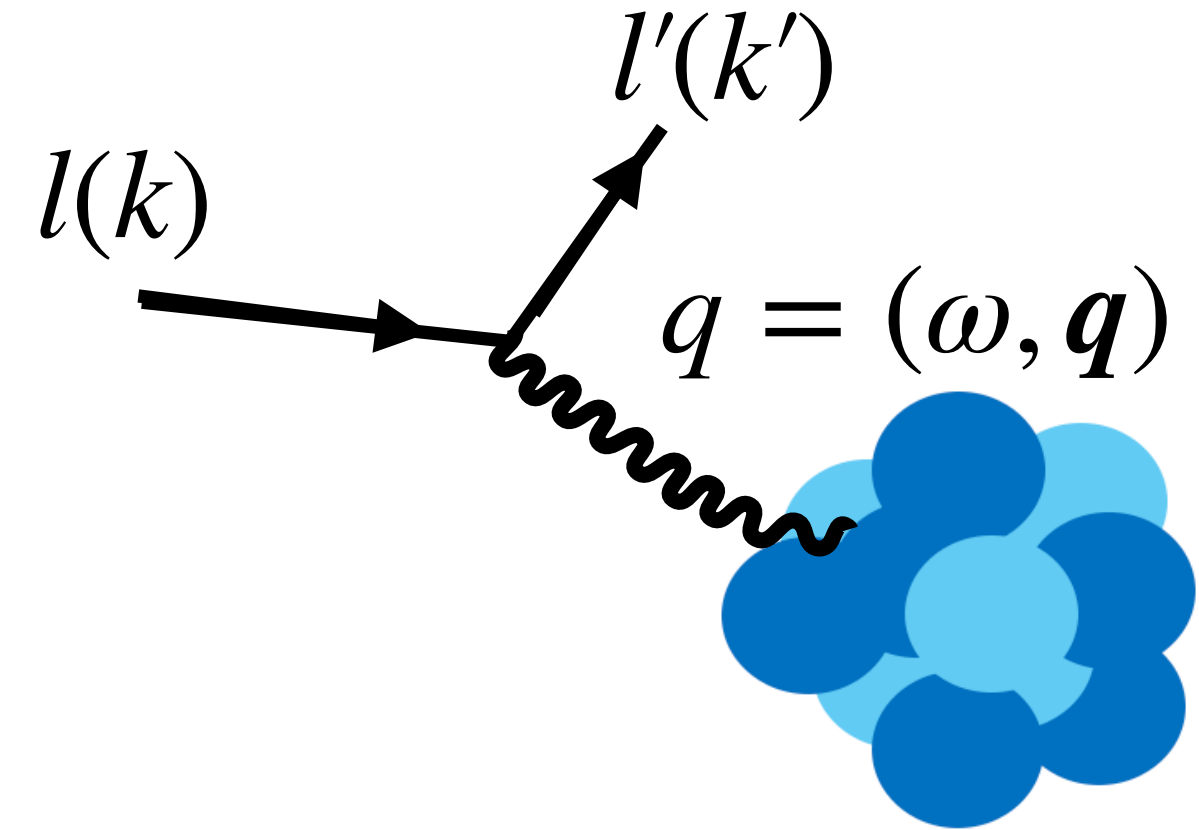
$$\frac{d^2\sigma}{dE'_k d\Omega'_k} = \kappa \frac{|k'|}{|k|} L_{\mu\nu}(k, k') W^{\mu\nu}(q)$$



Theoretical formalism

Inclusive lepton-nucleus cross section

$$\frac{d^2\sigma}{dE'_k d\Omega'_k} = \kappa \frac{|k'|}{|k|} L_{\mu\nu}(k, k') W^{\mu\nu}(q)$$



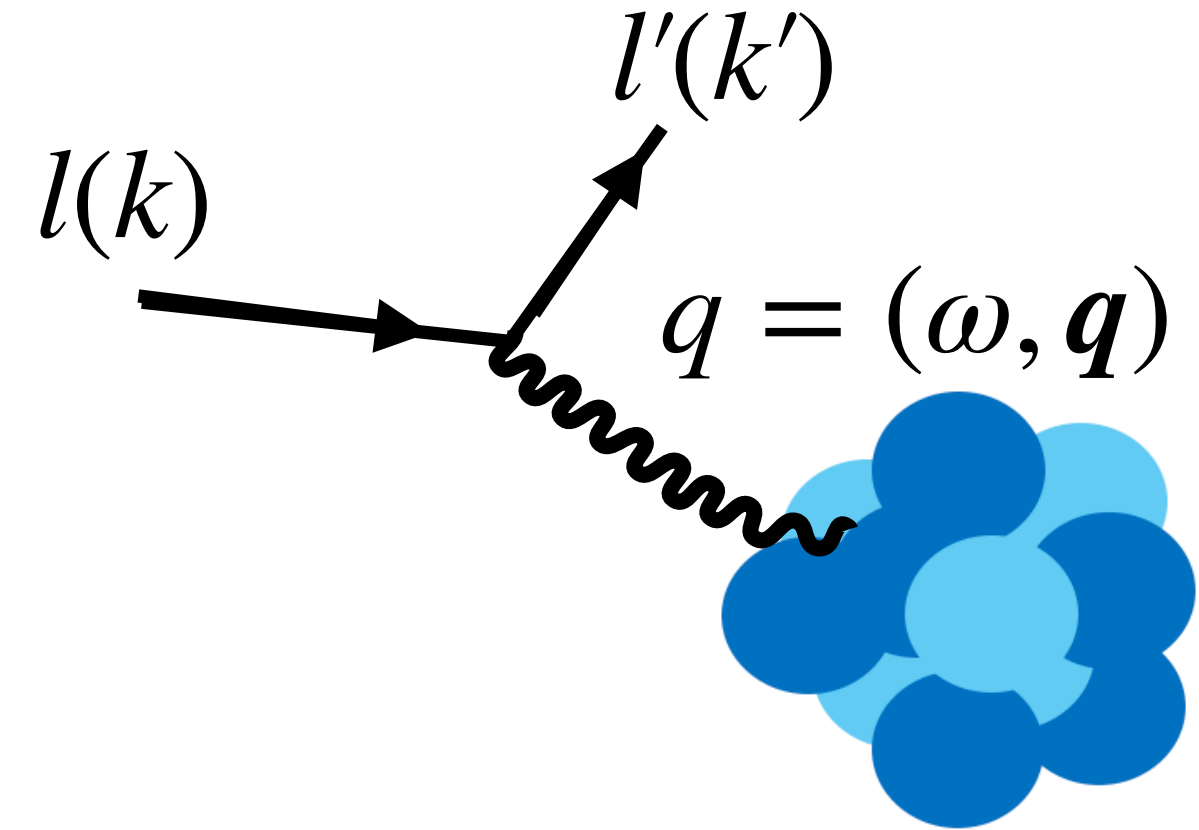
Impulse approximation (IA)

$$W^{\mu\nu}(q) = \int \frac{d\mathbf{p}}{(2\pi)^3} \int dE S_h(E, |\mathbf{p}|) S_p(E + \omega, |\mathbf{p} + \mathbf{q}|) \frac{1}{4E_p E_{p+q}} w^{\mu\nu}(p, q)$$

Theoretical formalism

Inclusive lepton-nucleus cross section

$$\frac{d^2\sigma}{dE'_k d\Omega'_k} = \kappa \frac{|k'|}{|k|} L_{\mu\nu}(k, k') W^{\mu\nu}(q)$$



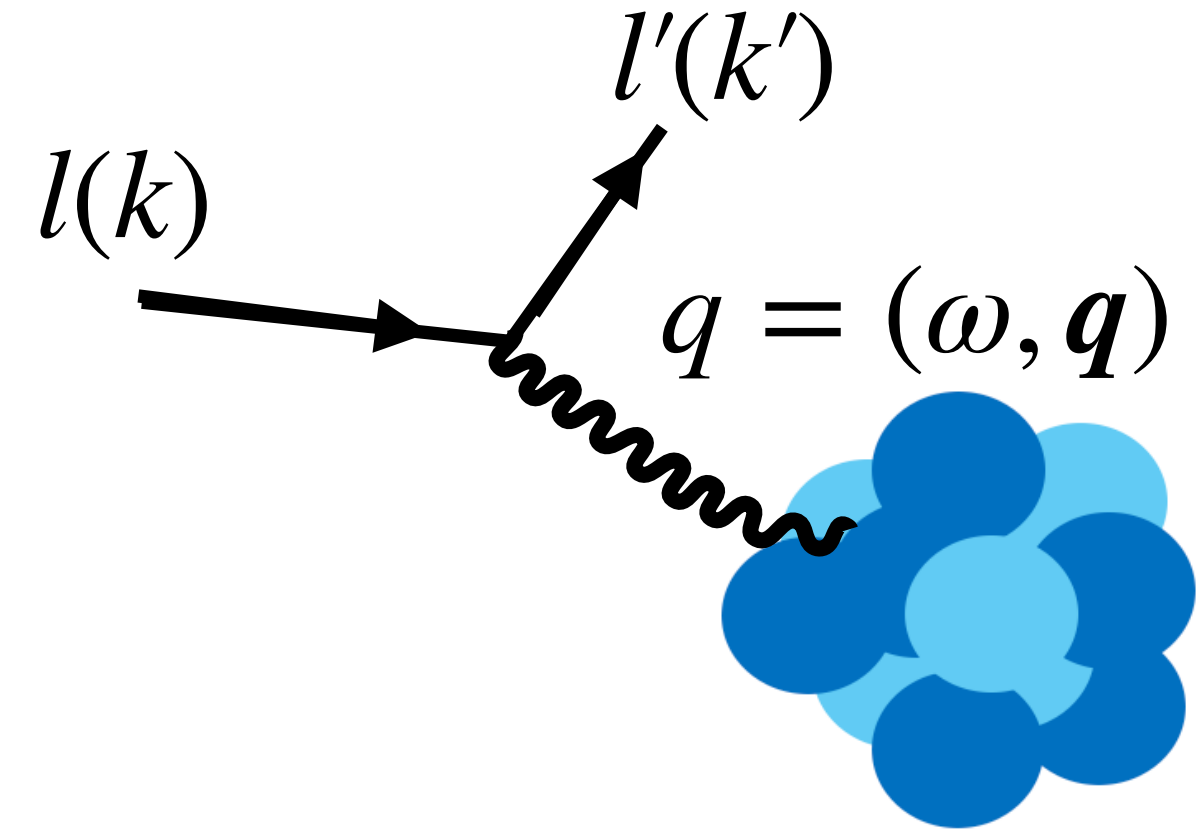
Impulse approximation (IA) + local density approximation (LDA)

$$W^{\mu\nu}(q) = \int d\mathbf{r} \int \frac{d\mathbf{p}}{(2\pi)^3} \int dE S_h^{NM}(E, |\mathbf{p}|, \rho(r)) S_p^{NM}(E + \omega, |\mathbf{p} + \mathbf{q}|, \rho(r)) \frac{1}{4E_p E_{p+q}} w^{\mu\nu}(p, q)$$

Theoretical formalism

Inclusive lepton-nucleus cross section

$$\frac{d^2\sigma}{dE'_k d\Omega'_k} = \kappa \frac{|k'|}{|k|} L_{\mu\nu}(k, k') W^{\mu\nu}(q)$$



Impulse approximation (IA) + local density approximation (LDA)

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Final-state interactions (FSI)

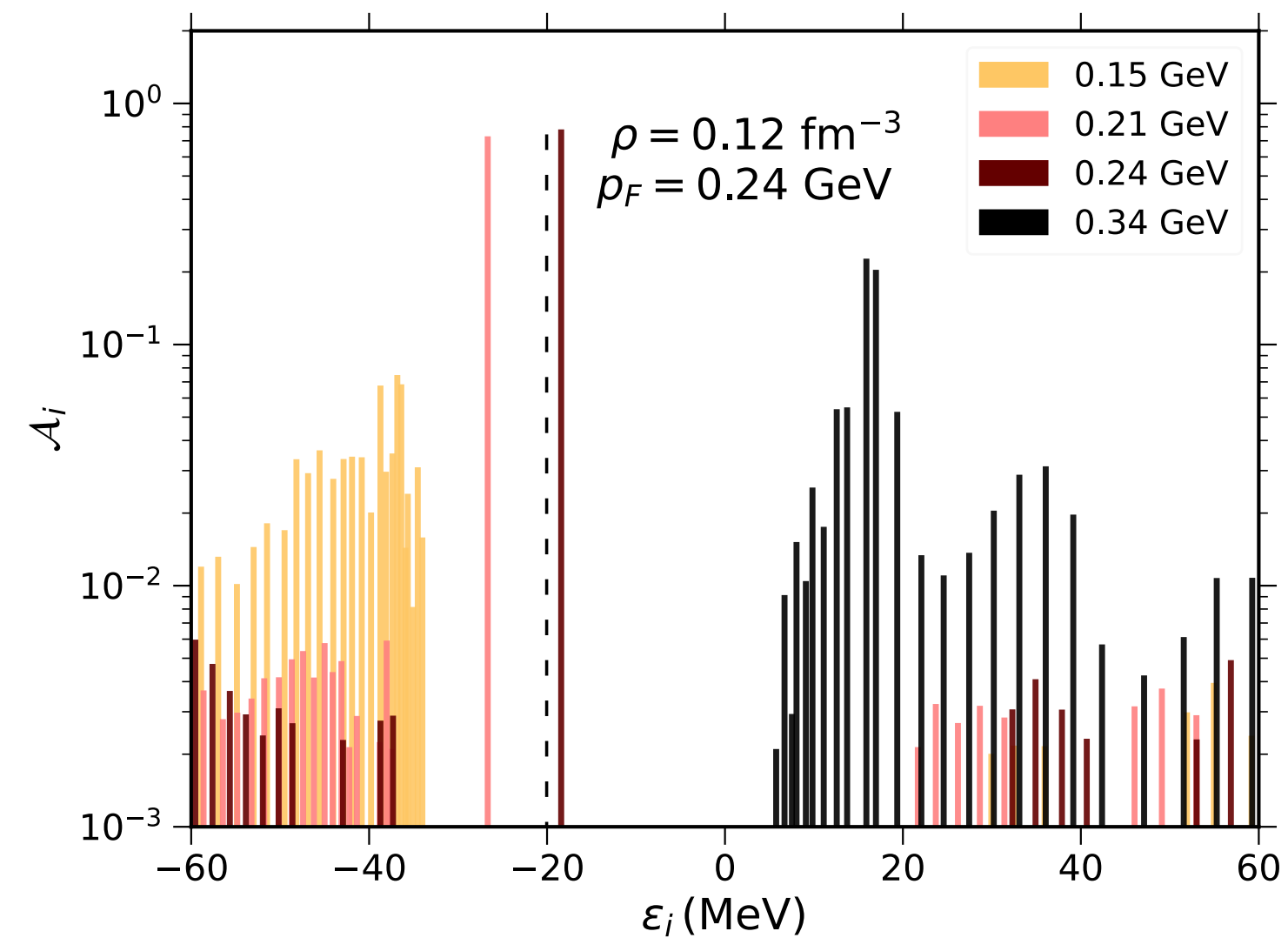
PWIA: S_h^{NM} from *ab initio*, S_p^{NM} from local relativistic Fermi gas

IA+FSI: S_h^{NM} and S_p^{NM} from *ab initio*

Neural-network interpolation of SFs

**Training data: discrete SF data
from symmetric nuclear matter**

$$S^{NM}(E, |\mathbf{p}|, \rho) = \sum_i \mathcal{A}_i \delta(E - \epsilon_i) \delta_{|\mathbf{p}|, p_i} \delta_{\rho, \rho_i}$$



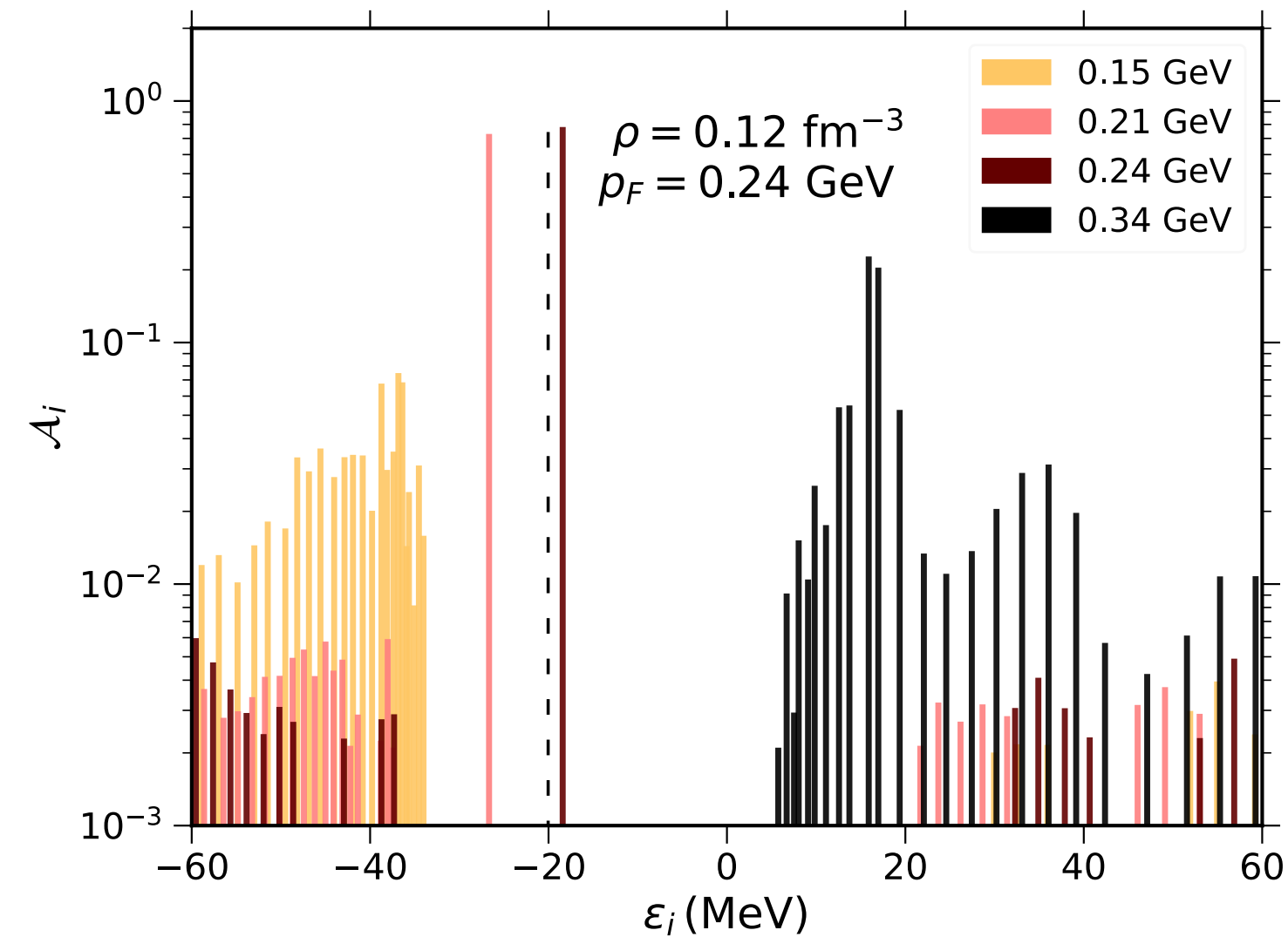
Neural-network interpolation of SFs

**Training data: discrete SF data
from symmetric nuclear matter**

Smoothing along E-dimension

$$S^{NM}(E, |\mathbf{p}|, \rho, \Gamma) = \frac{\Gamma}{\pi} \sum_i \frac{\mathcal{A}_i}{(E - \epsilon_i)^2 + \Gamma^2} \delta_{|\mathbf{p}|, p_i} \delta_{\rho, \rho_i}$$

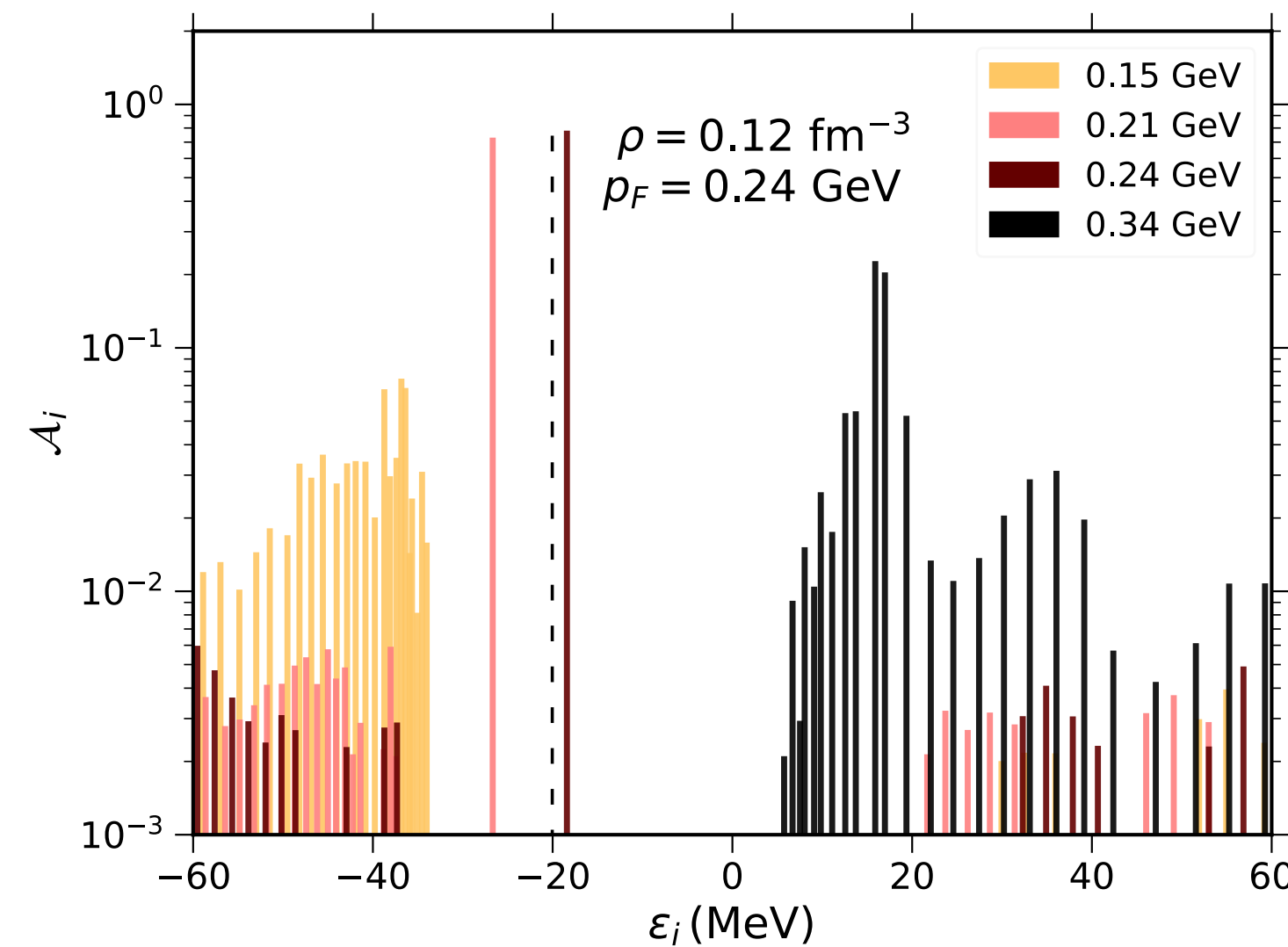
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Neural-network interpolation of SFs

Training data: discrete SF data from symmetric nuclear matter

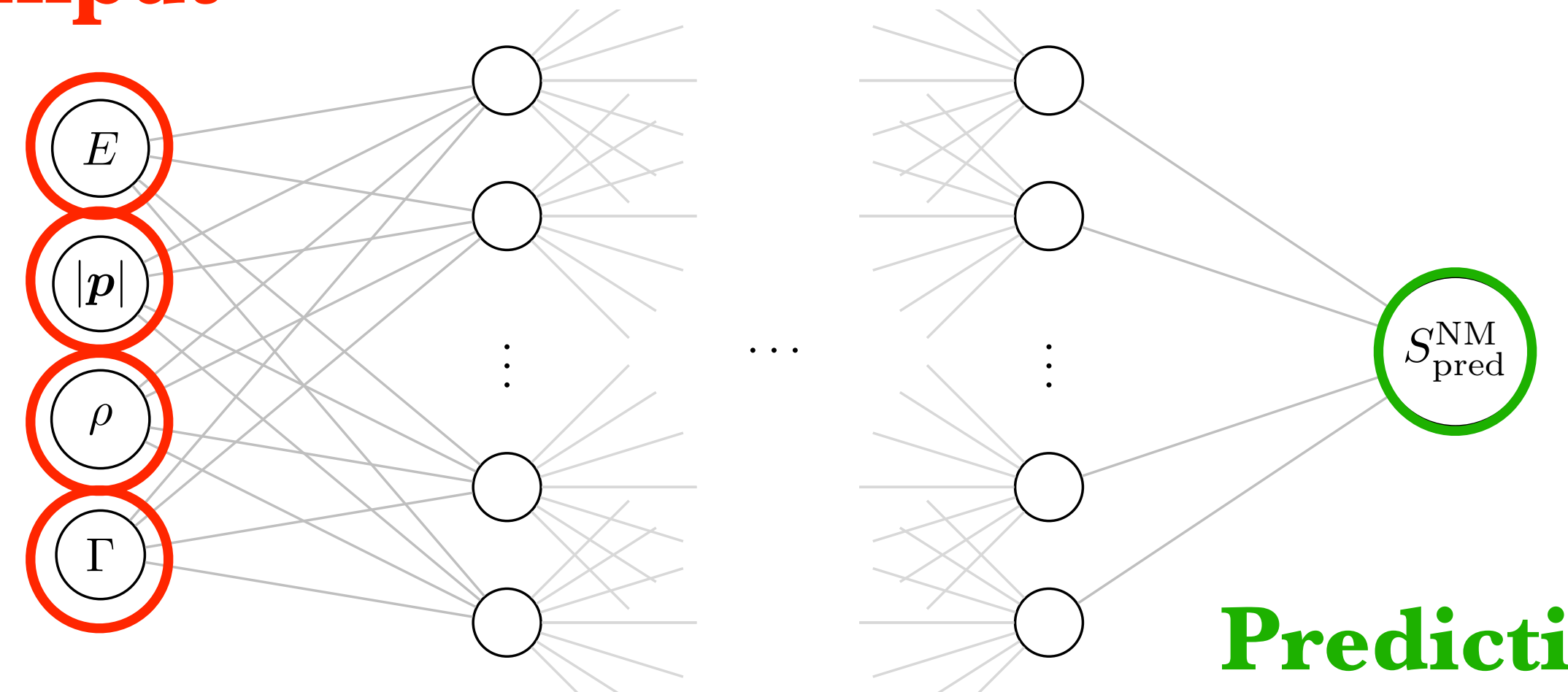
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Smoothing along E-dimension

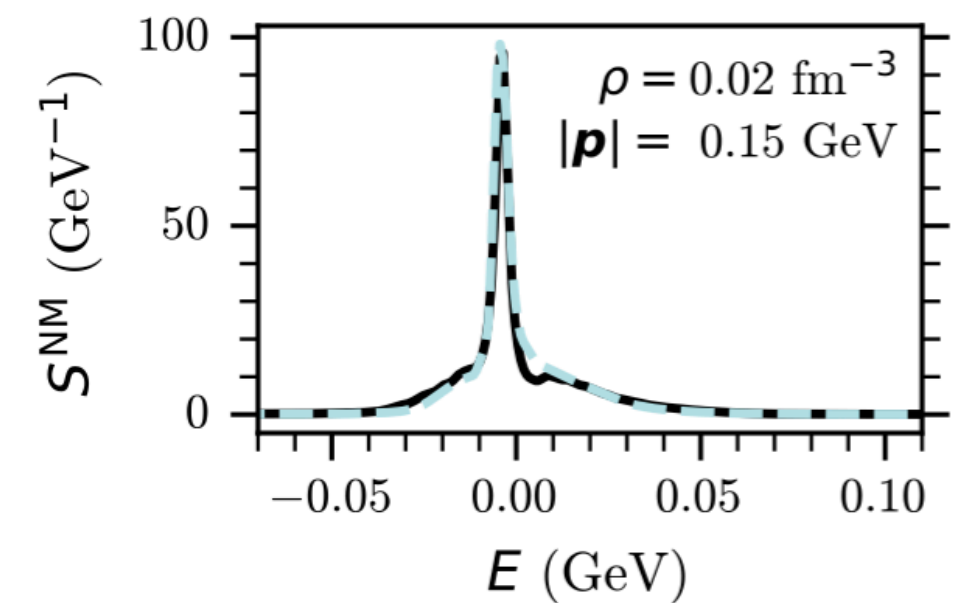
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Input

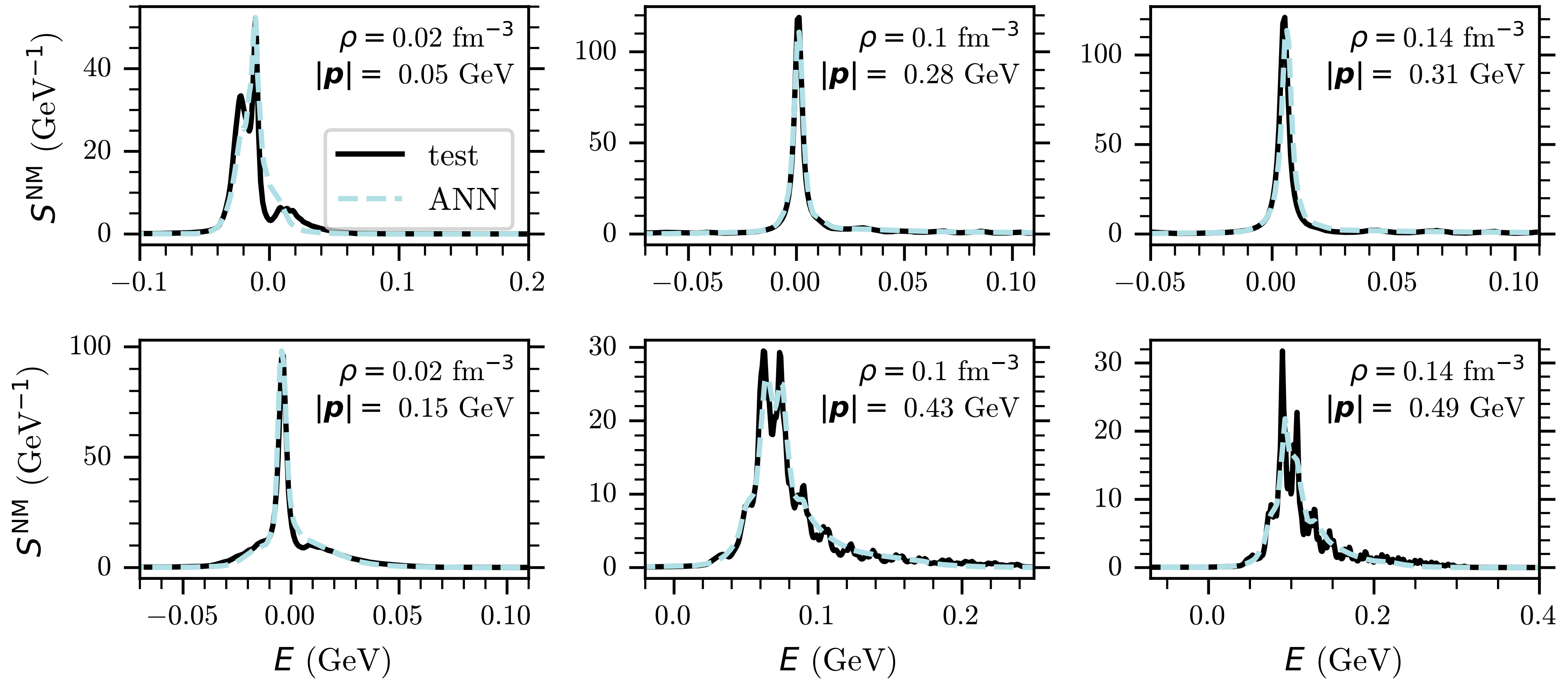


Prediction

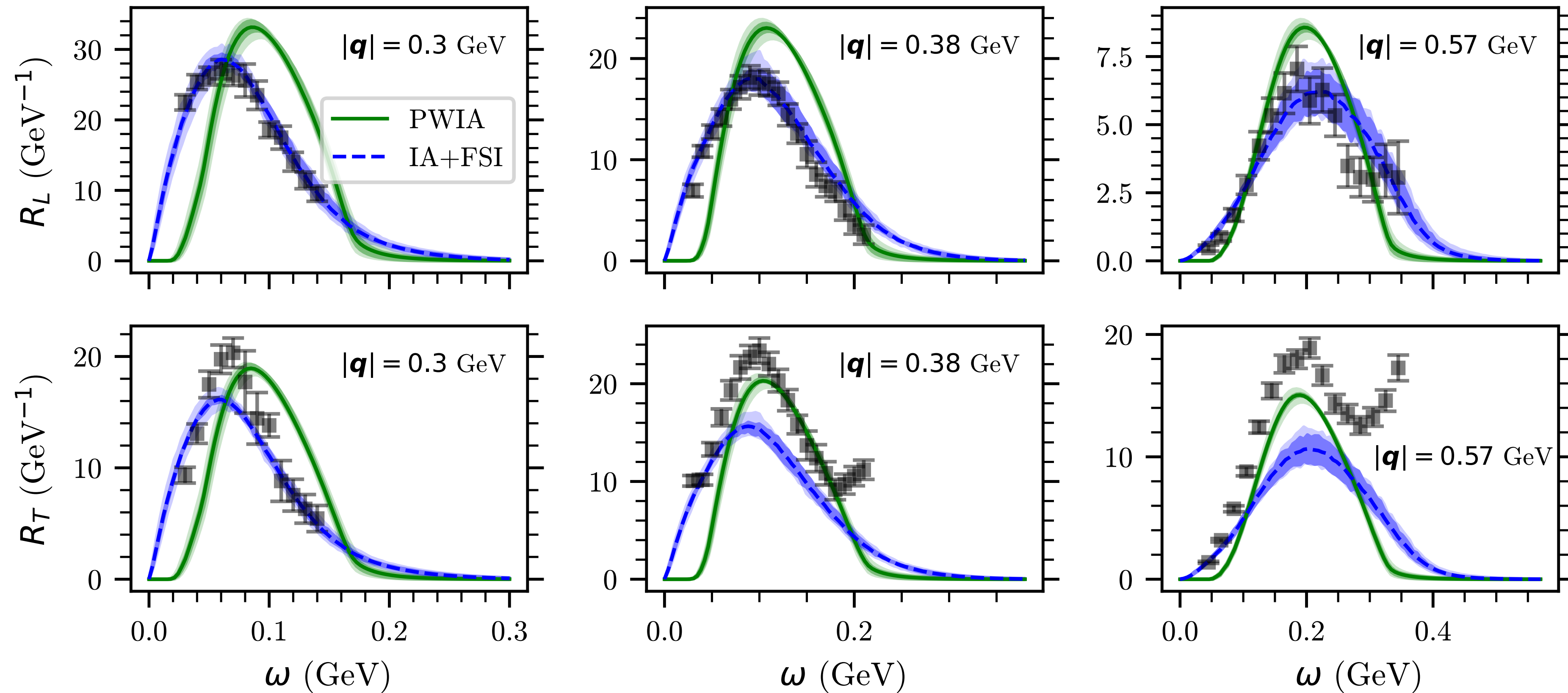
Three chiral Hamiltonians
 $\Delta N^2\text{LO}_{\text{GO}}(394)$ $\Delta N^2\text{LO}_{\text{GO}}(450)$
 $N^2\text{LO}_{\text{sat}}(450)$



Neural-network predictions

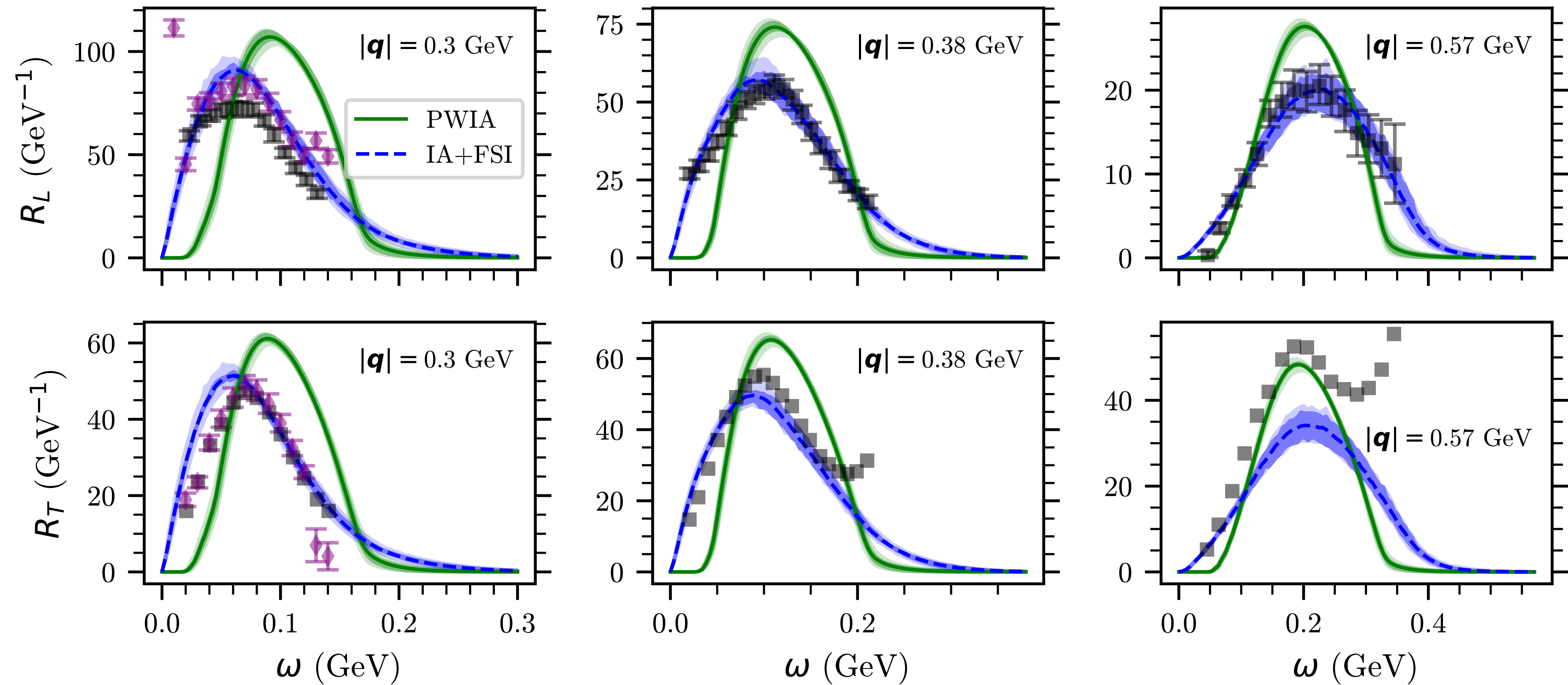


Electromagnetic response functions for ^{12}C



Experimental data from J. Jourdan (1996)

Electromagnetic response functions for ^{40}Ca



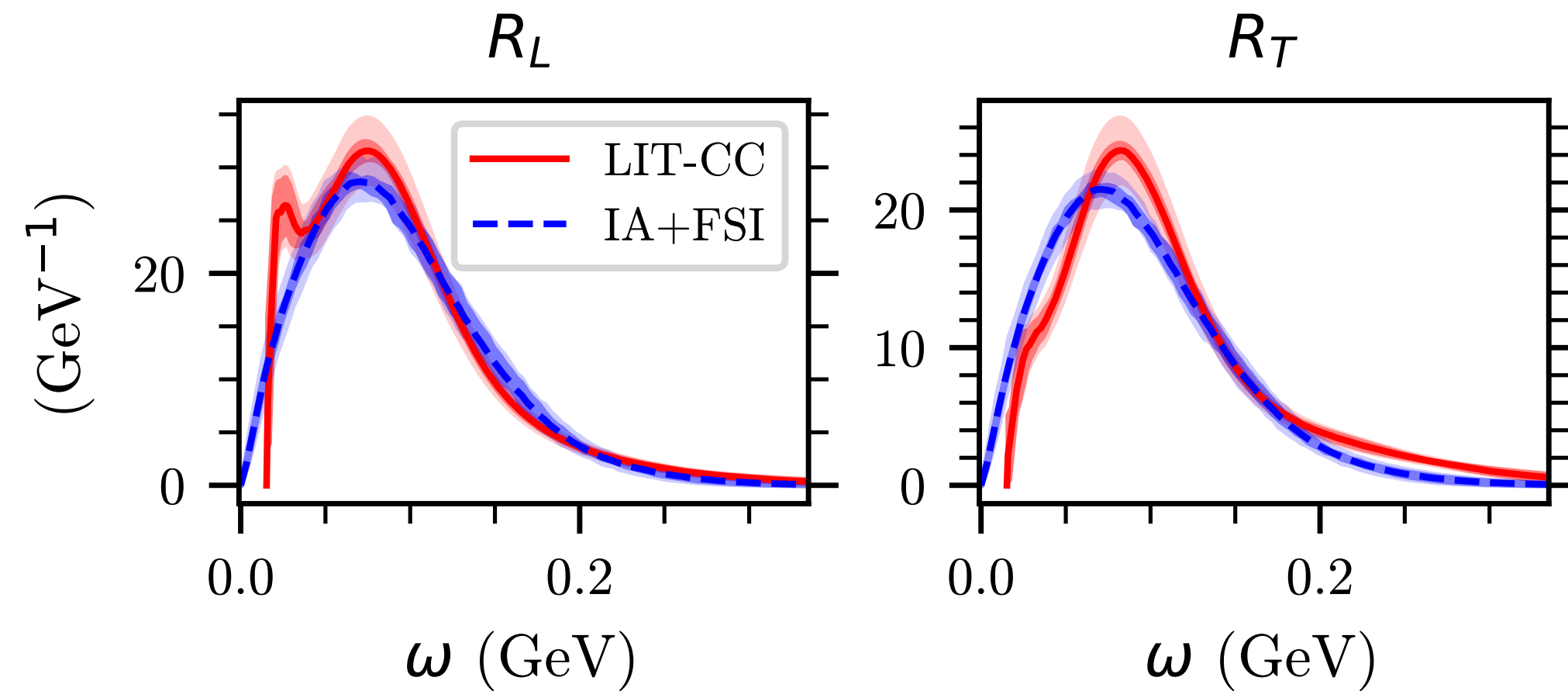
Experimental data from **W. Williamson *et al.* (1997)**, **J. Jourdan (1996)**

Response functions for ^{16}O

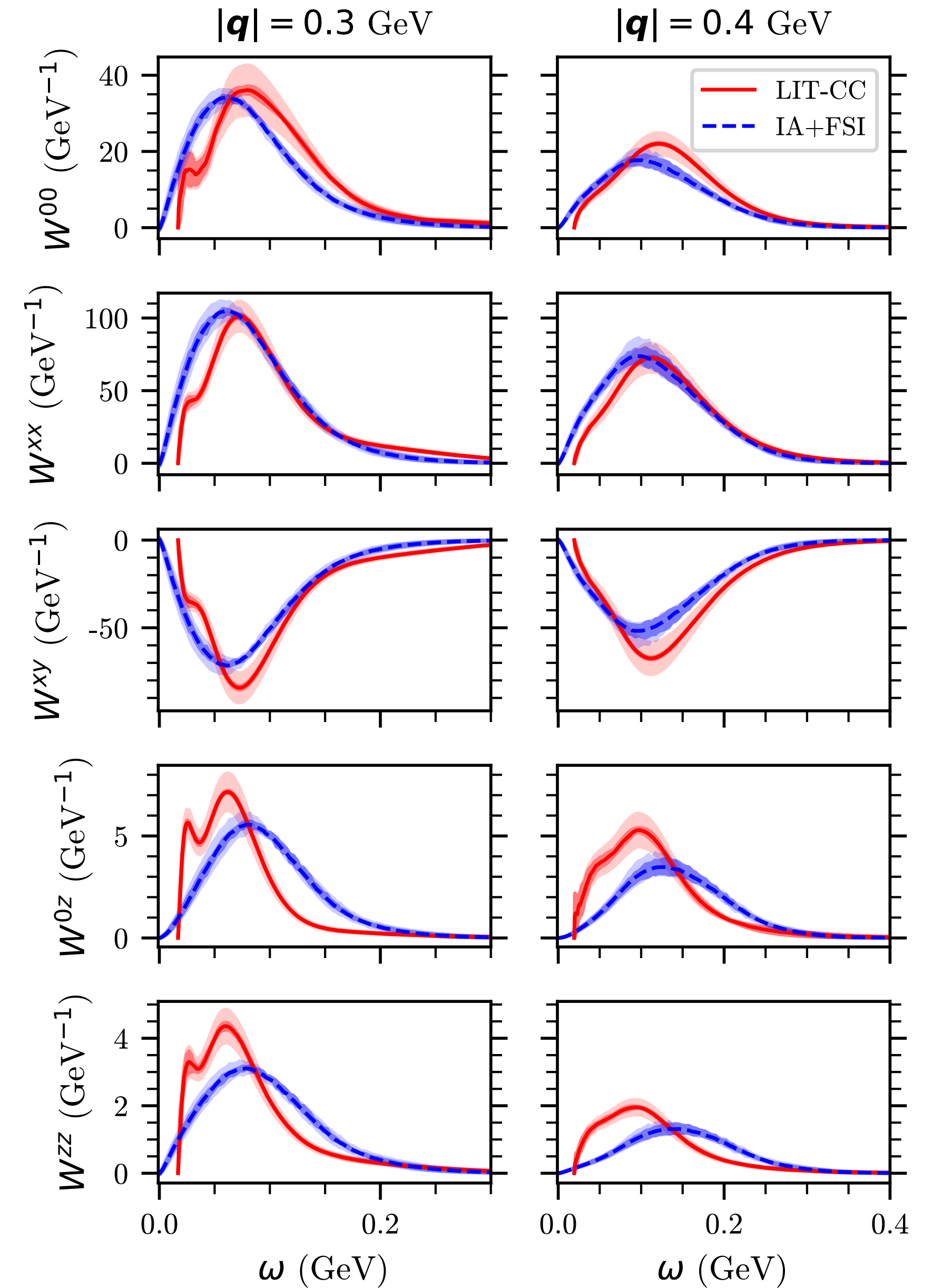
Charged current $\rightarrow$

Electromagnetic $\downarrow$

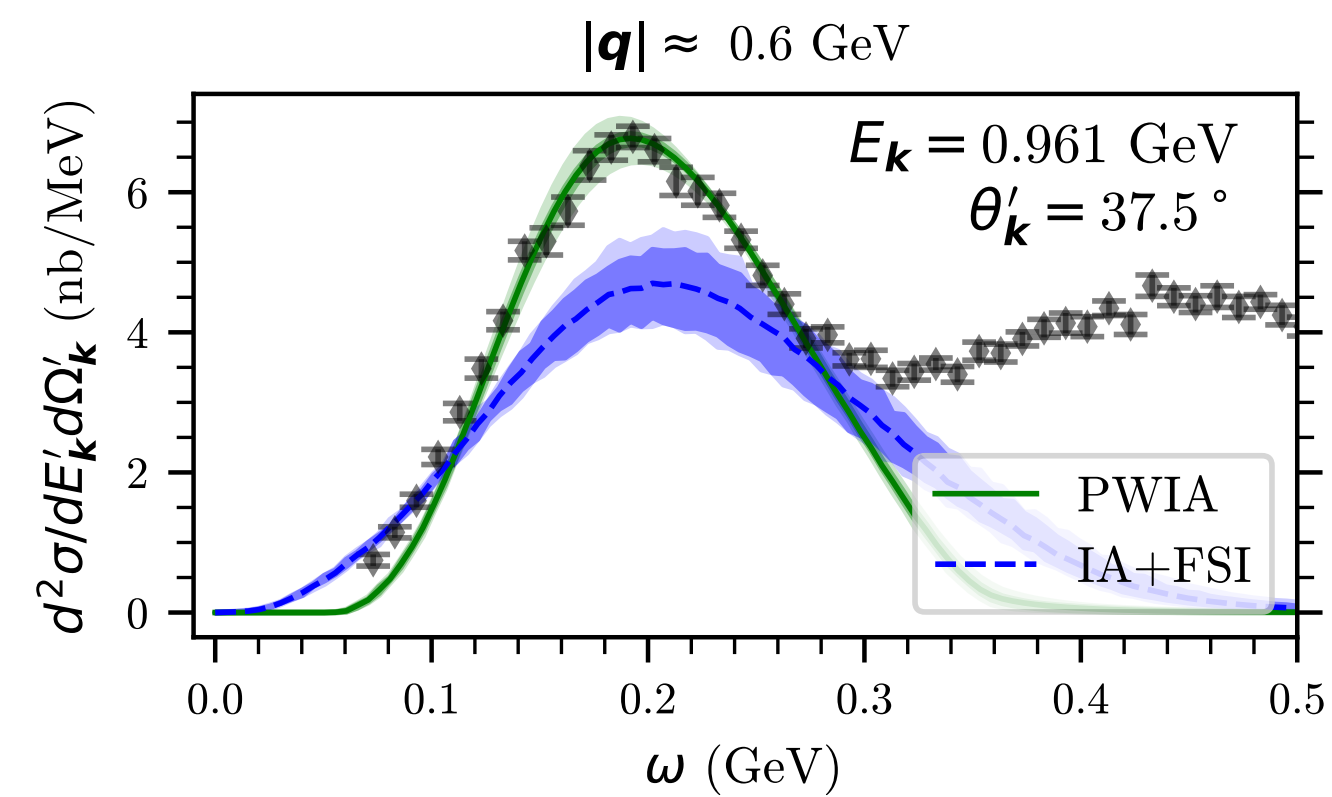
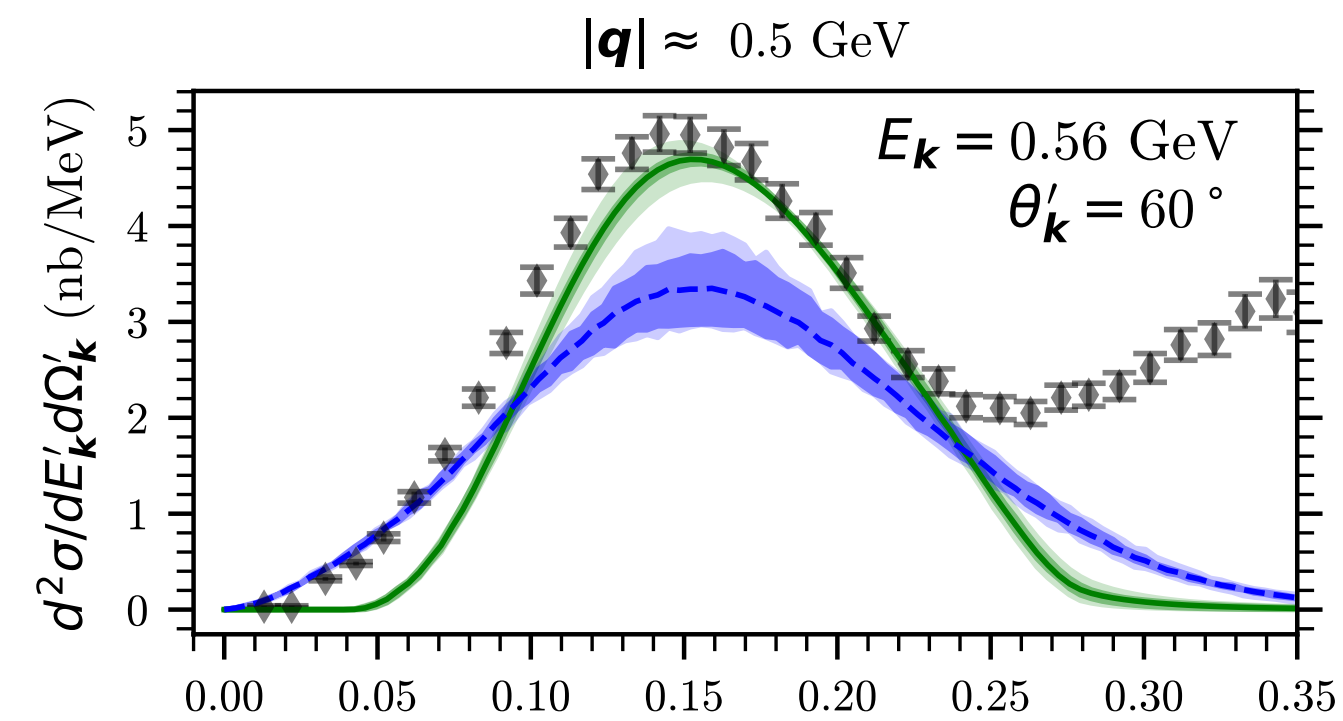
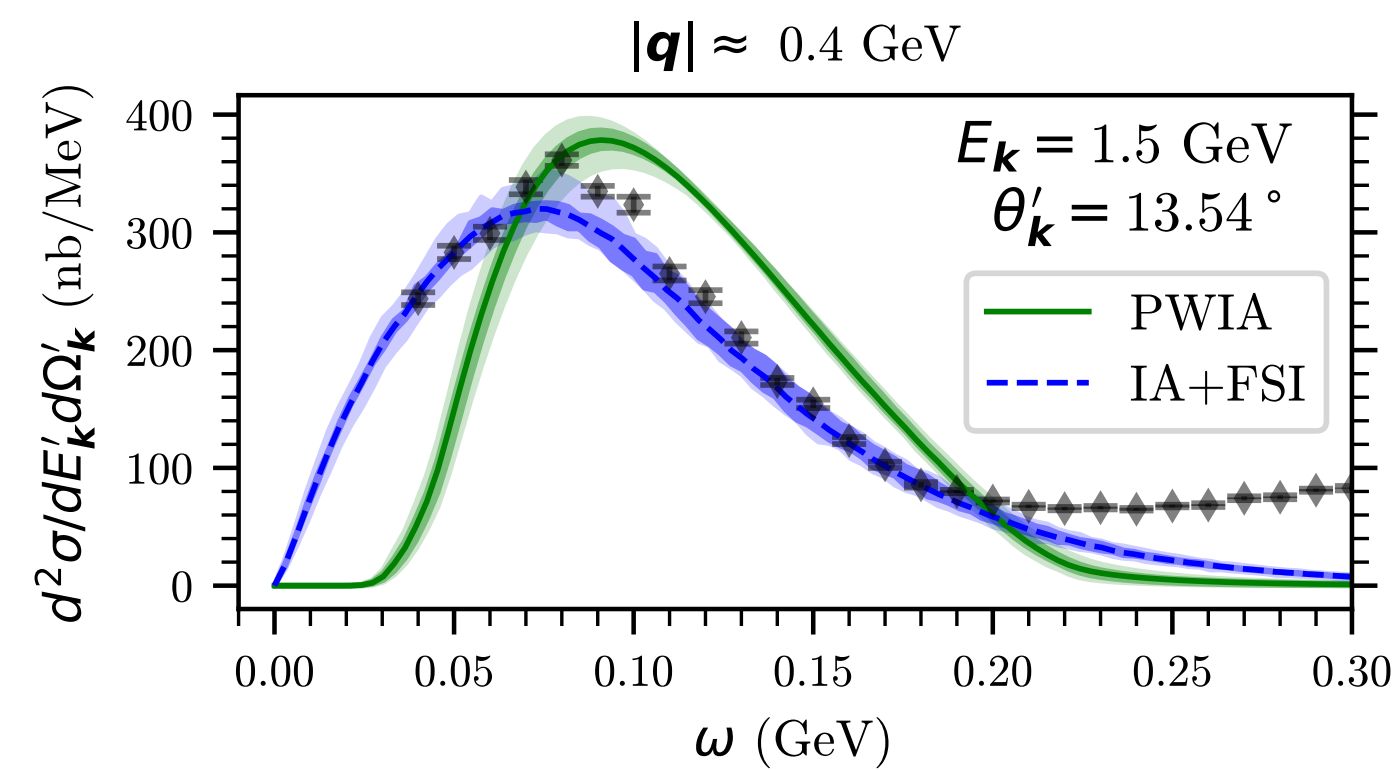
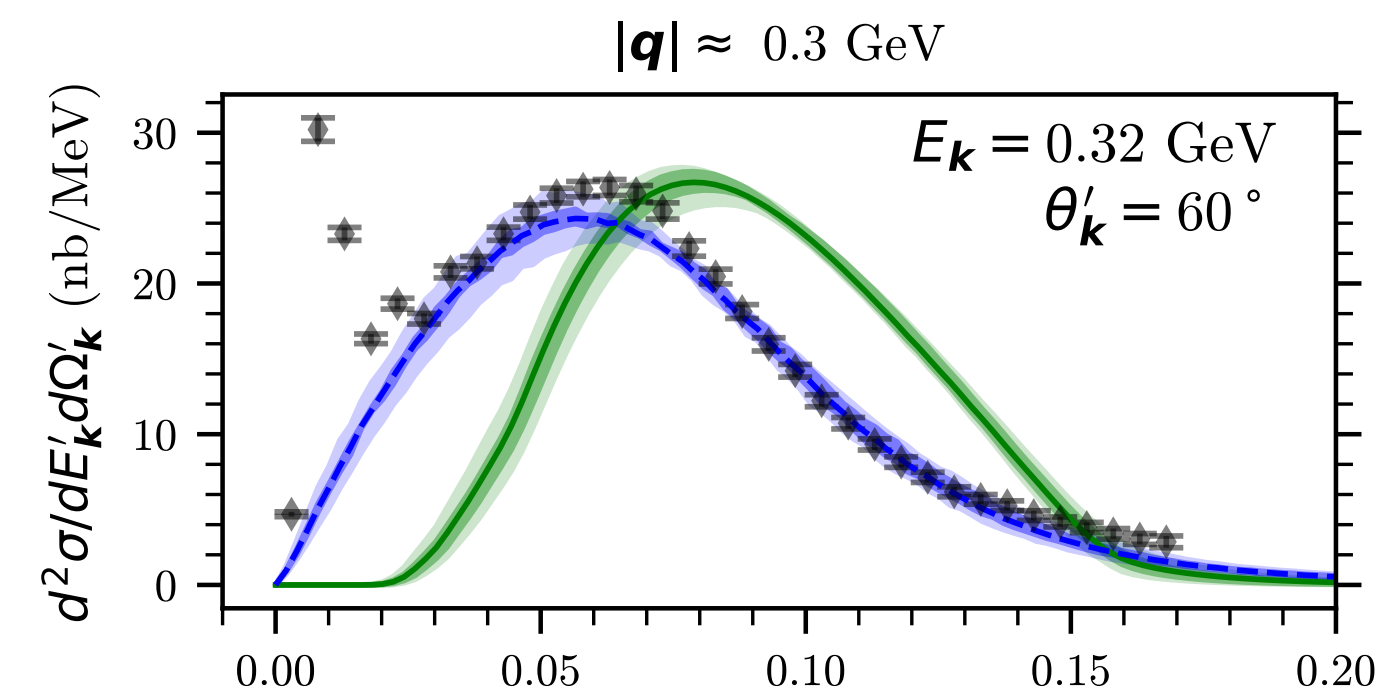
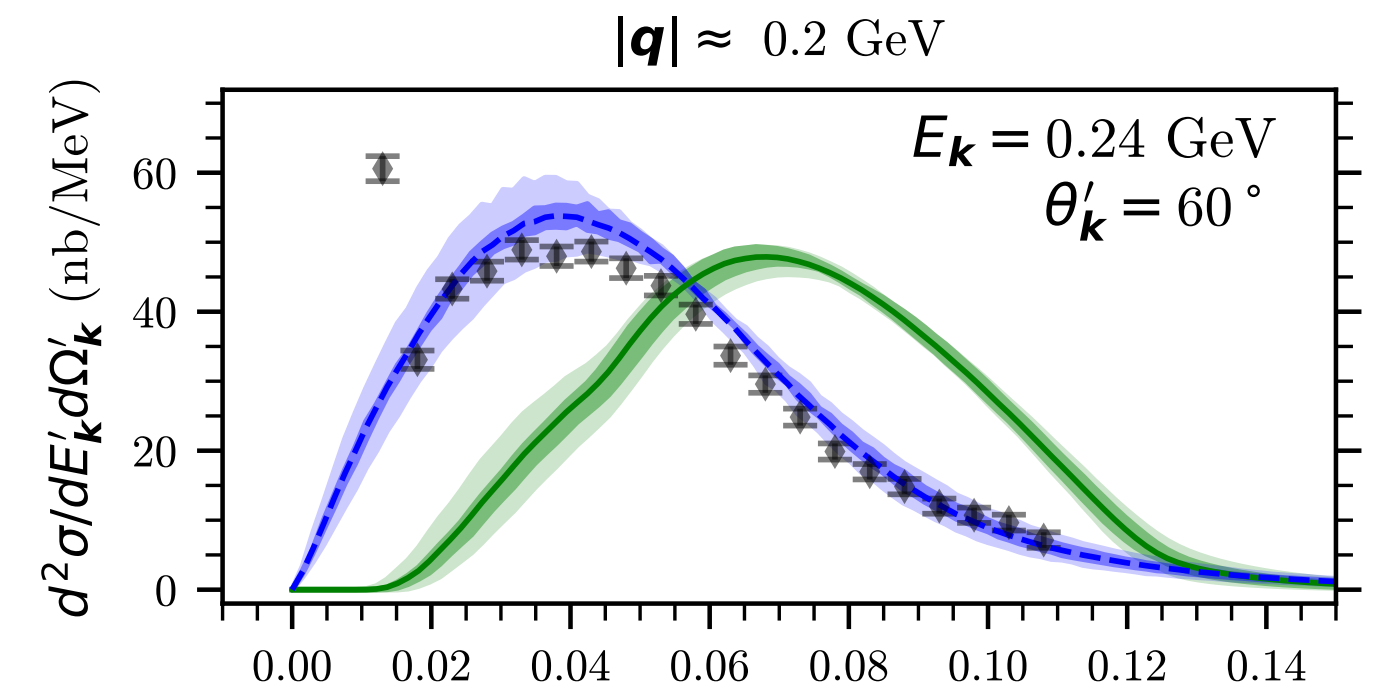
$|q| = 0.335 \text{ GeV}$



LIT-CC responses from B. Acharya *et al.* (2025)



Electron-¹²C inclusive cross section



Experimental data from O. Benhar *et al.* (2006)

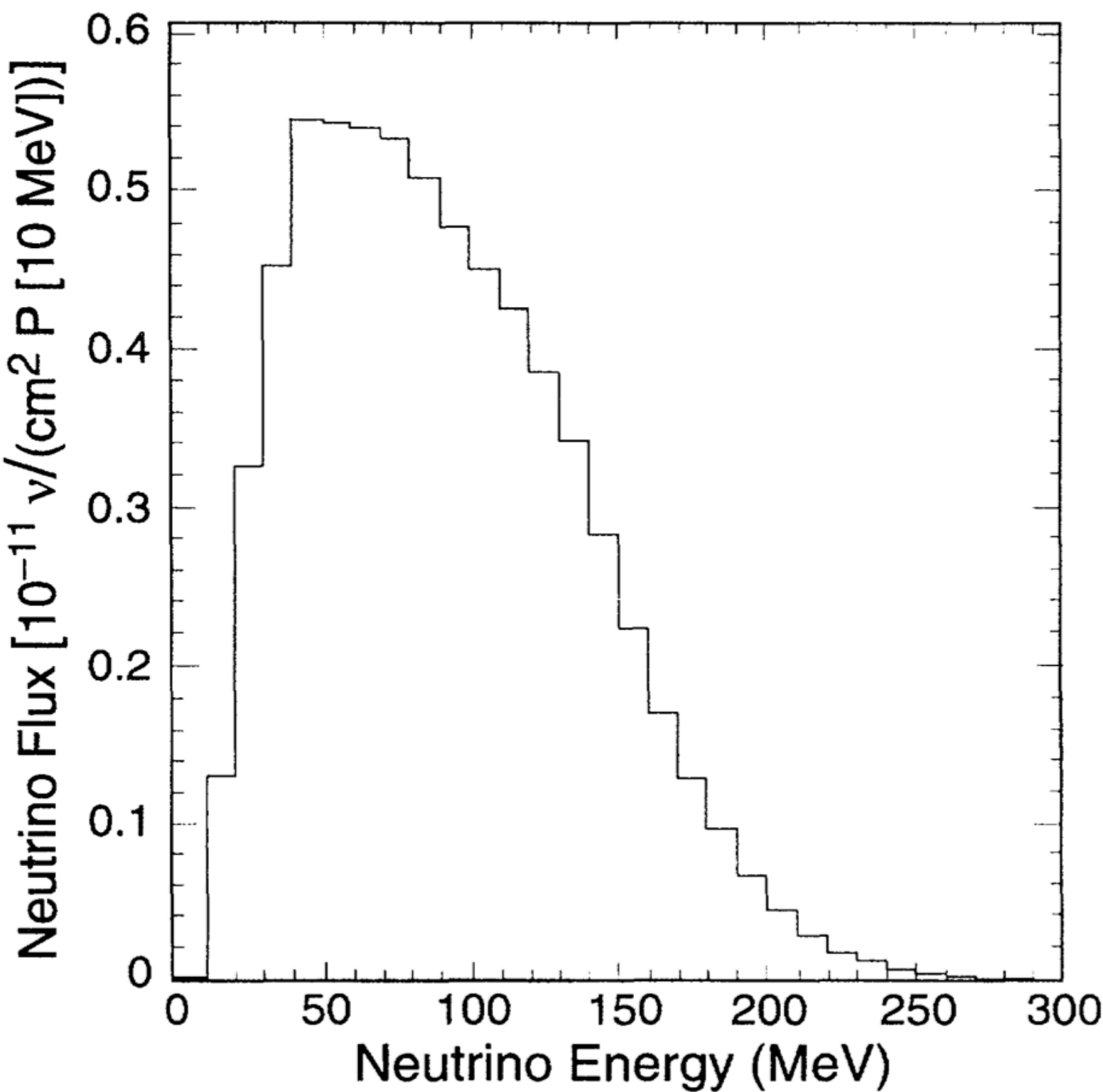
Flux-averaged total ν_μ - ^{12}C cross section $\bar{\sigma}_\mu$

$\bar{\sigma}_\mu$ in units of 10^{-40} cm^2

	PWIA	IA+FSI
$\Delta\text{N}^2\text{LO}_{\text{GO}}(394)$	7.99 ± 0.33	10.43 ± 0.08
$\Delta\text{N}^2\text{LO}_{\text{GO}}(450)$	8.31 ± 0.28	10.35 ± 0.12
$\text{N}^2\text{LO}_{\text{sat}}(450)$	7.97 ± 0.29	10.40 ± 0.16
Experiment		
LSND [72]	LSND [73]	LSND [74]
$8.3 \pm 0.7 \pm 1.6$	$11.2 \pm 0.3 \pm 1.8$	$10.6 \pm 0.3 \pm 1.8$

[72]: M. Albert *et al.* (LSND) (1995)
[73]: C. Athanassopoulos *et al.* (LSND) (1997)
[74]: L. B. Auerbach *et al.* (LSND) (2002)

Neutrino flux



M. Albert *et al.* (1995)

Improvements and next steps

Possible improvements

Quantify neural-network uncertainties
(Bayesian neural network)

Include EFT truncation error, not only the
uncertainty from varying Hamiltonian

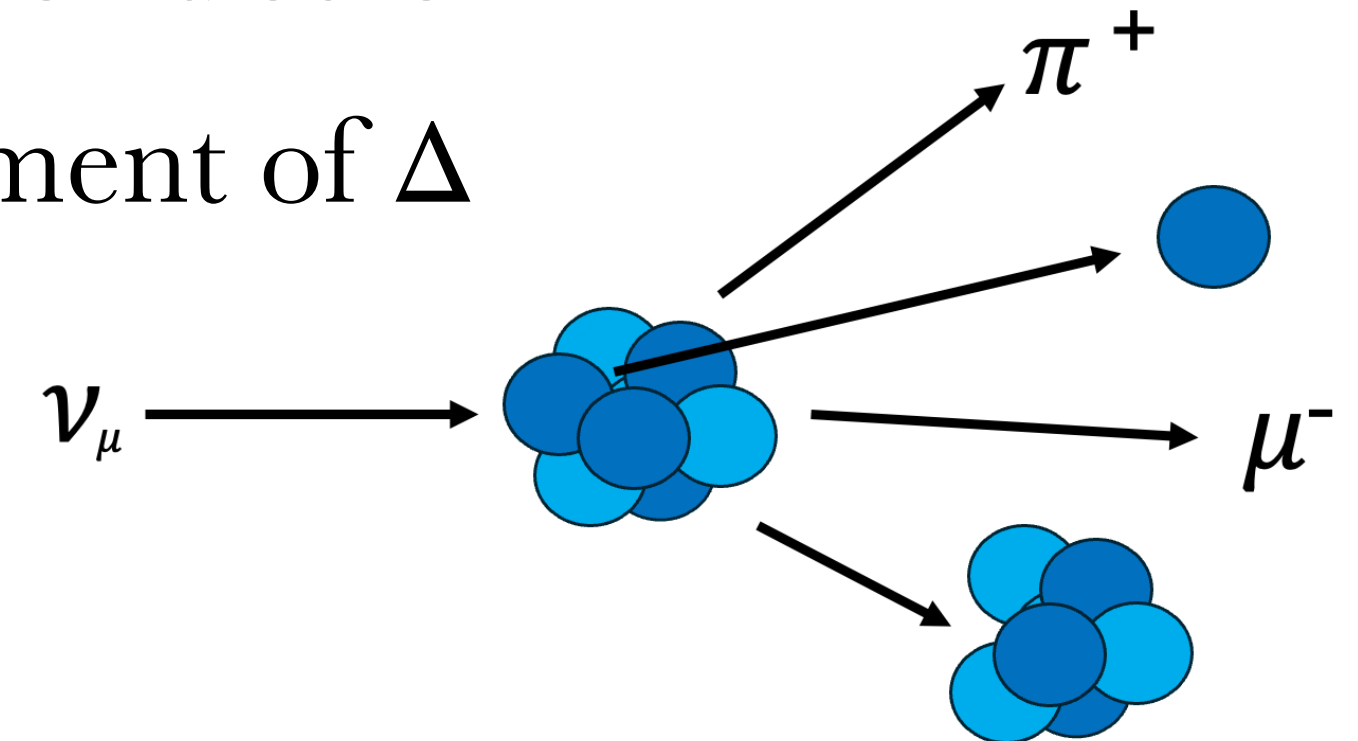
Better understanding of discrepancy between
LIT-CC and IA+FSI in CC responses

Strength at $\omega < S$ (separation energy)

Next steps

Single-pion production

In-medium treatment of Δ



Implementation into MC event generators?

Intra-nuclear cascade does not modify outgoing
lepton kinematics, but the particle SF does...

? Avoid double counting

Summary

Methods

- *Ab initio* FSI in the spectral function formalism (particle SF)
- Artificial neural networks as interpolators of the multidimensional SFs

Results

- Benchmark against inclusive QE electromagnetic data for several symmetric nuclei
- Clear improvement at low $|\mathbf{q}|$ for the IA+FSI model, compared to PWIA

Outlook

- Extension of framework to resonance region (in-medium Δ and single-pion production)
- Implementation into MC event generators?